

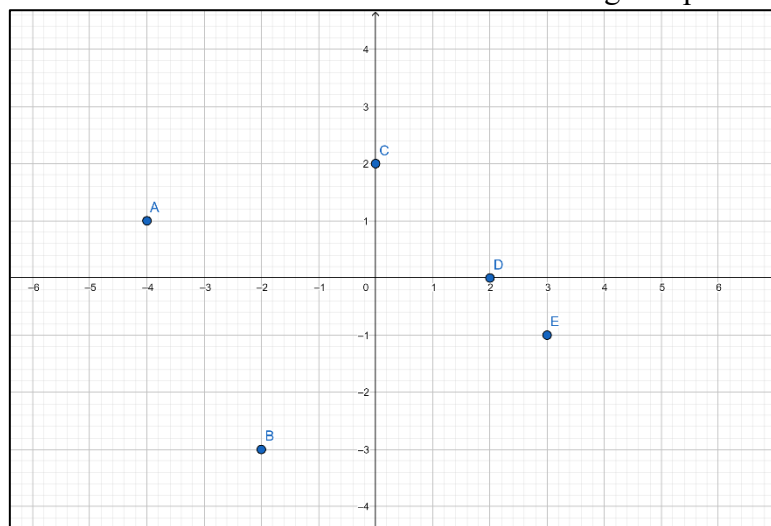
DPEN021: Enabling Mathematics A

Class Questions Week 4

Week 4A

Key Idea - Graphing Straight Lines

1. Use the graph below to write the coordinates of each of the given points.



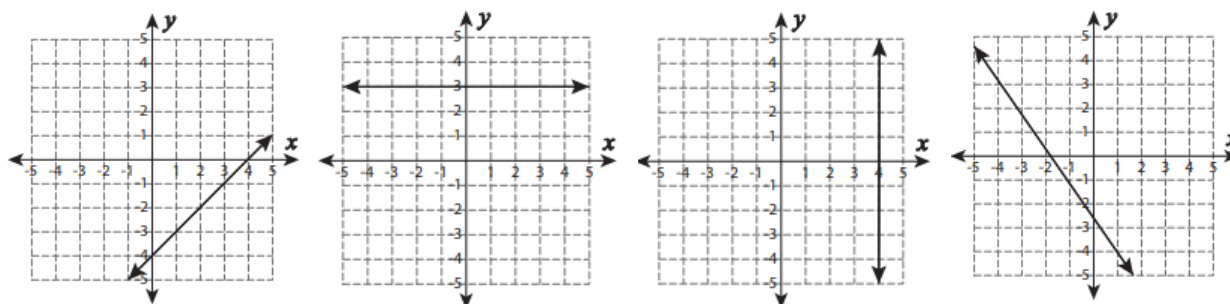
2. By drawing up tables of values (choose $x = -2, -1, 0, 1, 2$) plot the graphs of the following:
- (a) $y = \frac{1}{2}x + 2$ (b) $y = 3x - 1$ (c) $y = 2 - 3x$
- (d) $y + 2x + 2 = 0$ (e) $y + \frac{1}{3}x - 3 = 0$ (f) $y - x - 5 = 0$
3. By finding the x and y intercepts, plot a straight line graph for each of the following equations.
- (a) $y = 4 - 2x$ (b) $2x - y - 6 = 0$ (c) $\frac{2x + y}{3} = 1$
- (d) $\frac{x - y}{2} = 1$ (e) $2x + 3y = 12$ (f) $\frac{5x - 2y}{3} = -4$
4. Determine whether each point lies on the given line:
- (a) $(3, 5)$ on $y = 2x - 1$ (b) $(-1, 7)$ on $y = -3x + 1$ (c) $(0, -4)$ on $2x + y = -4$
5. Find the value of k so that the point $(k, 2)$ lies on the line $3x - y = 4$.
6. If the point $(1, m)$ lies on the line $y = -4x + 6$, find the value of m .

Week 4B

Key Idea - Properties of Linear Graphs

1. Given the following linear equations determine: (i) the slope (ii) x -intercept (iii) y -intercept
- (a) $y = -3x + 6$ (b) $3y = 7x$ (c) $y = \frac{3}{2}x - 1$
- (d) $3x + y - 12 = 0$ (e) $12x + 3y = 9$ (f) $y = \frac{5x - 6}{4}$

2. State whether each of the following lines has a positive gradient, negative gradient, or neither.



3. The vertices of a triangle are the points $A(2, -4)$, $B(5, 2)$, $C(1, -2)$. Find the slope of each side.
4. Find the gradient between the following pairs of points:
- (a) $(1, 2)$ and $(4, 8)$ (b) $(3, -1)$ and $(-1, 5)$ (c) $(-2, -2)$ and $(1, 4)$
 (d) $(0, 2)$ and $(-7, 1)$ (e) $(0, 10)$ and $(5, 10)$ (f) $(-2, -1)$ and $(-3, -2)$
5. For each pair of lines, decide if they are parallel, perpendicular or neither.
- (a) $y = 4x + 1$ (b) $y - 5x + 1 = 0$ (c) $y + 3x - 9 = 0$
 $2y = 8x - 5$ $5y + x + 10 = 0$ $2y - 6x + 2 = 0$
6. State the gradient of the line that is perpendicular to a line with a gradient of:
- (a) $m = 2$ (b) $m = -\frac{1}{4}$ (c) $m = \frac{3}{2}$
7. Show that the line joining $(-2, 3)$ & $(2, -5)$ is parallel to the line $y = -2x + 3$.
8. Show that the line joining $(6, 3)$ & $(9, 5)$ is perpendicular to the line joining $(5, 7)$ & $(9, 1)$.

Key Idea - Equations of Straight Lines

1. Find the equation of the straight line given the following conditions, write your answer in:
- (i) gradient-intercept form (ii) general form
- (a) Gradient $\frac{1}{3}$, y -intercept -3 (b) Gradient -3 , y -intercept 1
 (c) Passing through $(3, -1)$, gradient $m = -2$ (d) Passing through $(2, 3)$, gradient $m = 1$
 (e) Passing through the origin, slope of -4 (f) Passing through points $(4, 2)$, $(3, 1)$
 (g) Passing through points $(5, -1)$, $(2, -2)$
 (h) Passing through $(1, -5)$ and is parallel to the x -axis
 (i) Parallel to $y = 2x + 3$ with y -intercept 1
 (j) Parallel to $y = -3 + \frac{1}{2}x$ and passing through $(1, 3)$
 (k) Perpendicular to $x - 3y = 4$ with y -intercept 3
 (l) Perpendicular to $y = -3x - 2$ and passing through $(2, -8)$

Week 4C

Key Idea - Intersection of Two Lines & Simultaneous Equations

1. Solve the following simultaneous equations:

(a) $x + 2y = 8$ $2x + 3y = 13$	(b) $x + y = 10$ $x - y = 2$	(c) $\frac{3}{2}x - 4y = 7$ $x + \frac{1}{2}y = \frac{3}{2}$
(d) $3x - y = 5$ $5x + 3y = -8$	(e) $-2x + 7y = 4$ $-3x + 5y = -5$	(f) $5x + 2y = 9$ $9x - 7y = -5$
(g) $-3x + 6y = -9$ $-5x + 4y = -15$	(h) $\frac{3}{5}x + \frac{4}{5}y = 1$ $-\frac{1}{4}x + \frac{3}{8}y = 1$	(i) $2x + 5y = -13$ $2x - 3y = -5$
(j) $2x + 3y = 19$ $4x - y = 3$	(k) $2x + y = 11$ $x + y = 8$	(l) $3x + 2y = 16$ $x + 2y = 10$

- Find the equation of the straight line passing through the point of intersection of the lines $2x + 5y - 19 = 0$ and $3x - 4y + 6 = 0$ and has a gradient of 4. Give your answer in general form.
- Find the equation of the straight line passing through the point of intersection of the lines $4x + y + 3 = 0$ and $5x - 2y + 8 = 0$ and is parallel to the line with equation $3x - 2y + 5 = 0$. Give your answer in gradient-intercept form.
- Find the equation of the straight line passing through the point of intersection of the lines $3x + 2y - 12 = 0$ and $5x - y - 7 = 0$ and passes through the point $(-4, -5)$. Give your answer in general form.
- Find the equation of the straight line passing through the point of intersection of the lines $4x - y = 7$ and $2x + y = 11$ and is perpendicular to the line $y = 5x$. Give your answer in gradient intercept form.

Key Idea - Function Notation

1. Identify the independent and dependent variables:

(a) $h = 2t + 9$	(b) $s = 5n - 1$	(c) $y = x^2 - 3x + 1$
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2. Given $f(x) = -x^2 - 4x + 2$, find:

(a) $f(3)$	(b) $f(-2)$	(c) $f(a+1)$
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3. Given $g(t) = 2t^2 - 3t$, find:

(a) $g(2x)$	(b) $g(x-1)$	(c) $g(a+b)$
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4. Given $F(x) = \frac{3x+2}{4x-1}$, find:

(a) $F(1)$	(b) $F(a-4)$	(c) $F\left(\frac{t}{2}\right)$
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5. (a) Given $f(x) = 2x + |x|$, find $f(-3) - f(5)$.

(b) Given $f(x) = 3x^2 - 3x + 2$ and $g(x) = 4x - 1$, find $f(x+h) - g(x)$.

(c) Given $f(x) = 3^{-x}$ $g(x) = 1 - x^2$, find $f(2) - g(-1)$.

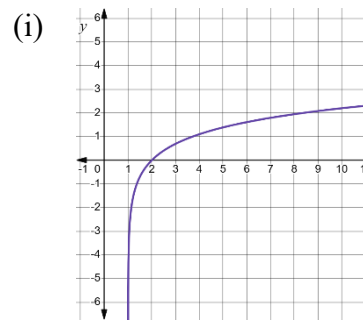
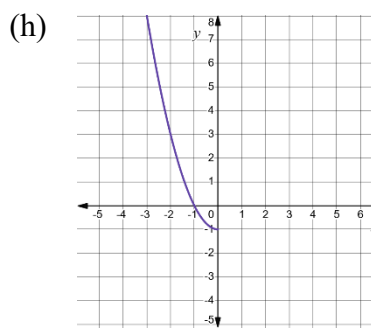
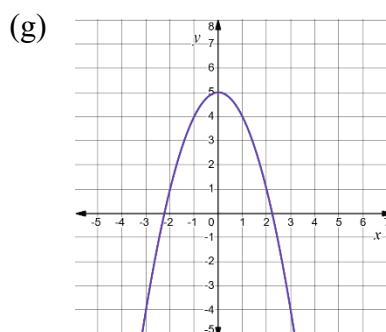
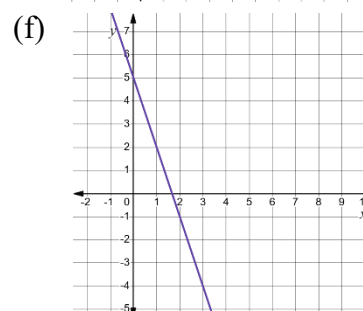
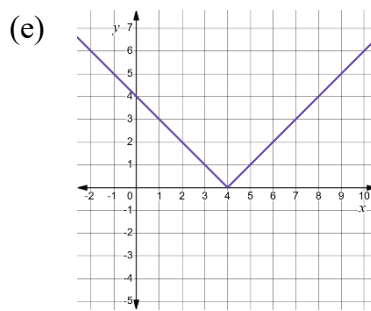
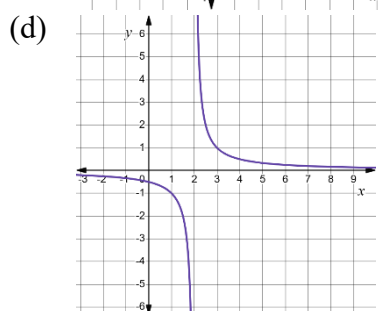
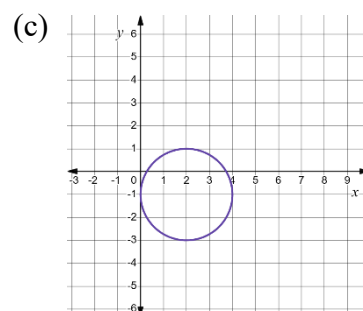
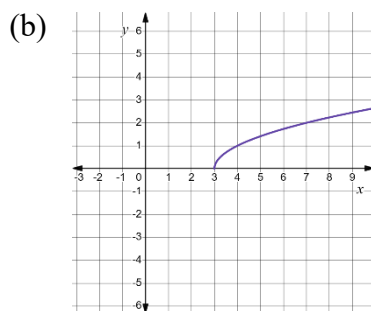
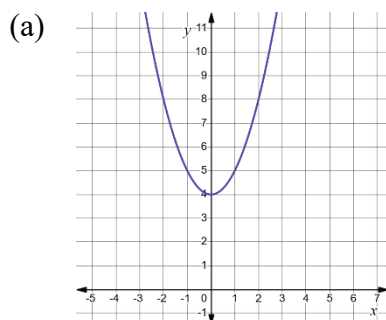
6. Find the roots to the following equations:

(a) $f(x) = 2 - 3x$	(b) $g(x) = x^2 - 9$	(c) $h(x) = -x^2 + 4x + 5$
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Week 4D

Key Idea – Finding Domain and Range Graphically

1. Find the domain and range of the following functions, write your answers in interval notation.



Key Idea – Finding Domain and Range Algebraically

1. Find the domain and range of the following functions, write your answers in interval notation.

(a) $y = 2x + 3$

(b) $y = -4x + 1$

(c) $y = \frac{1}{2}x - 5$

(d) $y = \sqrt{x - 4}$

(e) $y = \sqrt{x}$

(f) $y = \sqrt{6 - 2x}$

(g) $y = \frac{2}{3 - x}$

(h) $y = \frac{5}{2x + 1}$

(i) $y = \frac{1}{x}$

(j) $y = \frac{1}{\sqrt{x - 3}}$

(k) $y = \frac{3}{\sqrt{4 - 2x}}$

(l) $y = \frac{1}{\sqrt{9x - 3}}$

Key Idea – Piecemeal Functions

1. (a) A function is defined by $f(x) = \begin{cases} 6, & x < 3 \\ 2x, & x \geq 3 \end{cases}$, find $f(4)$ and $f(-2)$.
- (b) A function is defined by $f(x) = \begin{cases} 2x+1, & x \geq 2 \\ 9-2x, & x < 2 \end{cases}$, find $f(0)$, $f(2)$ and $f(4)$.
- (c) A function is defined by $g(t) = \begin{cases} 2t^2, & t \leq -1 \\ 1+2t, & -1 < t < 2 \\ t, & t \geq 2 \end{cases}$ find $2g(0) + \frac{1}{2}g(-1) - g(3)$.
- (d) A function is defined by $f(x) = \begin{cases} x^2, & x > 1 \\ 1-x^2, & x \leq 1 \end{cases}$, find $f(0) + f(2) - 2f(1)$.

Key Idea – Odd and Even Functions

1. Are these functions odd, even or neither?
- | | | |
|---------------------------------------|-----------------------------|---|
| (a) $f(x) = 4$ | (b) $f(x) = 2x$ | (c) $f(x) = 2x - 1$ |
| (d) $f(x) = x^5 - 4x^3 + x$ | (e) $f(x) = x^2 - 9$ | (f) $f(x) = 3x^4 + 2x$ |
| (g) $f(x) = \frac{3x}{x^2 - 3}$ | (h) $f(x) = x^6 - 3x^3 + 2$ | (i) $f(x) = \frac{2x}{x^3 + x}$ |
| (j) $f(x) = x^4 - 7x^2 + 3$ | (k) $f(x) = x(x^2 - 5)$ | (l) $f(x) = \frac{x^2 + 1}{x^2 - 4}$ |
| (m) $f(x) = \frac{x^3 - 2x}{x^2 + 1}$ | (n) $f(x) = \sqrt{x^2 + 4}$ | (o) $f(x) = \frac{x(x^2 - 1)}{x^2 - 9}$ |